

## PRACTICAL COURSE WORKBOOK

# Applied Prompt Engineering

Build a responsible AI workflow for a real work task

|                  |                                                                      |
|------------------|----------------------------------------------------------------------|
| LEVEL            | Intermediate                                                         |
| GUIDED TIME      | 5 hours across 12 lessons                                            |
| PROJECT          | a documented prompt, verification and revision workflow              |
| SKILLS           | Prompting, Chain-of-thought, RAG, Critical evaluation, Documentation |
| EDITORIAL STATUS | Structured draft v0.9                                                |

## WHAT YOU WILL BE ABLE TO DO

- Use Prompting appropriately in a realistic, bounded task.
- Use Chain-of-thought appropriately in a realistic, bounded task.
- Use RAG appropriately in a realistic, bounded task.

## WHO THIS IS FOR

People who want dependable, responsible AI-assisted workflows.

## BEFORE YOU BEGIN

- Basic computer and web skills

---

This open workbook is a structured draft undergoing course-specific editorial review. It does not provide certification.

## COURSE MAP

# From foundation to finished work

Applied Prompt Engineering is a structured, practical course covering Prompting, Chain-of-thought, RAG. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

| MODULE                                      | LESSONS AND FOCUS                                                                                           |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>01 Prompting: Foundations</b>            | 1. Capabilities with Prompting<br>2. Limitations with Chain-of-thought<br>3. Responsible use with RAG       |
| <b>02 Chain-of-thought: Working methods</b> | 1. Clear instructions with Prompting<br>2. Context design with Chain-of-thought<br>3. Verification with RAG |
| <b>03 RAG: Applied workflows</b>            | 1. Research with Prompting<br>2. Creation with Chain-of-thought<br>3. Automation with RAG                   |
| <b>04 Prompting: Quality and review</b>     | 1. Evaluation with Prompting<br>2. Privacy with Chain-of-thought<br>3. Repeatable practice with RAG         |

## HOW TO STUDY EACH LESSON

- 1 Read the objective and identify the decision it supports.
- 2 Reproduce the worked example with the stated input.
- 3 Test one normal case and one boundary or failure case.
- 4 Complete the knowledge check without guessing.
- 5 Save a short evidence note: result, limitation and next action.

## CAPSTONE PROJECT

# Produce a reviewable Applied Prompt Engineering project using Prompting, Chain-of-thought, RAG.

|             |                                                                                                     |
|-------------|-----------------------------------------------------------------------------------------------------|
| DELIVERABLE | A working Applied Prompt Engineering artefact plus an evidence-based self-review.                   |
| TOOLS       | A suitable Prompting environment, A plain-text decision log, Test data or realistic sample material |

## Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using Prompting.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

## Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

## Evidence log

|            |                                                      |
|------------|------------------------------------------------------|
| INPUT      | Original brief, data, requirements or test material. |
| DECISIONS  | Chosen method and one reasonable alternative.        |
| TESTS      | Normal, boundary and failure results.                |
| LIMITATION | What the result does not establish.                  |
| REVISION   | One evidence-led improvement.                        |

## REVIEW AND SOURCES

# Make the work credible and repeatable

## Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

## Authoritative references

### AI Risk Management Framework 1.0

NIST - reviewed 2026-08-21

<https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-ai-rmf-10>

### AI Principles

OECD - reviewed 2026-08-21

<https://www.oecd.org/en/topics/ai-principles.html>

---

#### SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

#### NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Prompting scope.